Slow by Design: Exploring How Museum Apps Can Cultivate Sustained Attention

Imogen Drews

[1]Film University Babelsberg KONRAD WOLF, Germany

Abstract

Digital applications in cultural institutions increasingly shape how visitors engage with exhibits, yet little is known about how their design can support sustained attention. This paper addresses this gap through a phenomenological auto-hermeneutic case study of the Jewish Museum Berlin application, applying Shari Tishman's slow looking strategies to Lennart Björneborn's framework of environmental affordances for serendipity. Focusing on the sub-affordance of slowability, the study examines how specific interface features foster slow looking, allowing visitors to observe, reflect, and discover details over extended periods. Findings reveal which design elements encourage or hinder sustained engagement, offering insights for intentional museum app design. This research contributes to understanding the intersection of digital design, attention, and cultural engagement, suggesting practical implications for museums navigating the attention economy.

1. Introduction

On a summer day in Berlin, I follow the Jewish Museum Berlin (JMB) website's suggestion to download its application. Choosing the Long Highlights Tour, intended to last ninety minutes, I find myself immersed for nearly two hours. My perception of time seems to stretch and contract, raising the question of how the app's design structured this sustained attention. The experience of engagement felt shaped not only by the museum itself but also by the rhythms and cues embedded in the app's interface. This quality of sustained immersion aligns with what education researcher Shari Tishman terms *slow looking* - a practice of "taking the time to carefully observe more than meets the eye at first glance" to reveal complexities that quick observation cannot capture [Tis18].

This paper presents a phenomenological auto-hermeneutic analysis of this experience, applying Lennart Björneborn's framework for enhancing serendipity [Bjö17]. This analysis particularly focuses on the sub-affordance of *slowability*, which captures how design can foster decelerated, attentive, and sustained forms of engagement [Bjö17]. While the concept has been discussed in relation to information environments more broadly, it has received little attention in the context of museum applications. The JMB was selected as a case study due to its accessibility, which facilitated the in-depth, iterative engagement required for an auto-hermeneutic analysis. It allowed for a sustained and reflective interaction with both the application and the physical exhibition space over multiple visits.

Through this analysis, the paper aims to identify which design features in a museum application like the JMB's foster the affordance of slowability. It also highlights the importance of intentionally designing for this, demonstrating its value not only for museums but for society at large.

2. Literature Review

2.1. Digital Mediation in Museums

Museums have increasingly shifted their focus from collecting and preserving artifacts to centering the visitor experience [Hei12]. With this change comes new challenges, particularly in the age of the *attention economy* [GLAAFR20]. Social media platforms, designed to maximize engagement through notifications, short-form content, and emotional triggers, have reshaped how people direct their focus [GLAAFR20]. One consequence for museums is that visitors spend less time with artworks than might be desirable, averaging just 27 seconds per piece [CP21].

In response, museums have explored various strategies to extend engagement. One such initiative is *Slow Art Day*, an annual event that invites visitors to spend about ten minutes observing a single artwork before sharing their reflections with others [CP21]. While this method has shown promise in group contexts, it is less clear whether digital tools can create the same depth of focus in individual experiences [GBdSVB21].

2.2. Designing for Serendipity: An Affordance Framework

This paper approaches that challenge through the lens of Lennart Björneborn's work on serendipity, which explores how spaces can be designed to allow for unplanned yet meaningful encounters [Bjö17]. These moments - when someone stumbles upon something unexpected that sparks curiosity - are, as he argues, both fundamental to culture and often underestimated. Similarly, the Victoria and Albert Museum highlights museums as particularly suited to fostering such encounters, aiming to provide visitors with personal, memorable, and sometimes surprising experiences [Gra22]. Achieving this requires environments (physical or digital) that encourage visitors to pause and linger with what they encounter.

Björneborn stresses that serendipity cannot be engineered directly but can be supported through affordances—conditions that make it more likely to occur. He identifies three: diversifiability (the variety of contents), traversability (how users move through or navigate an environment), and sensoriability (how an environment can be perceived through the senses) [Bjö17]. This paper focuses primarily on diversifiability and traversability, with particular attention to a sub-affordance of the latter: slowability. Slowability describes the extent to which an environment encourages slowing down, stopping, and paying closer attention. One way of fostering this, Björneborn suggests, is through *stumbling*—moments of friction or disruption, such as constraints or obstacles, that interrupt flow and prompt deeper engagement [Bjö17].

Designing for slowability can help museum-goers connect more deeply with artworks and notice details they might otherwise miss. In digital contexts, however, the challenge is sharper: users increasingly expect apps to run smoothly and without friction. The key question, then, is how to translate slowability into app design without undermining those expectations [GBdSVB21].

2.3. The Practice of 'Slow Looking'

Education researcher Shari Tishman's concept of slow looking offers a constructive alternative to the sometimes disruptive friction of stumbling. She defines it as an attentional practice that reveals complexity through extended observation [Tis18]. Although framed in terms of sight, she notes that learning can arise through all five senses. The aim is to deepen understanding of the world and uncover complexities that require more time to appreciate.

Tishman identifies several strategies for slow looking, drawn from science, art, and everyday practice [Tis18]:

- *Categories*: guiding attention in a particular direction, often used by museum educators.
- *Open inventories*: encouraging viewers to describe everything they notice, with the act of description itself revealing new details.
- *Scale and scope*: shifting physical perspective to see things differently.
- *Juxtaposition*: placing objects side by side to highlight contrasts.

This paper draws on Tishman's strategies to extend Björneborn's notion of slowability, with the aim of identifying app features that both sustain attention and enrich the visitor's experience.

3. Methodology and Case Study: The Jewish Museum Berlin App

3.1. Methodology: An Auto-Hermeneutic Phenomenological Approach

This study employs a phenomenological approach to examine the affordance of slowability. Data was collected through a structured auto-hermeneutic process in which I acted as the sole participant, systematically recording and analyzing my lived experience with the JMB app. The term auto-hermeneutics comes from information scholar Gorichanaz, who introduced it as a method for using one's own first-person experience to characterize a specific phenomenon [Gor17]. This approach differs from autoethnography in that it privileges deep interpretation of a phenomenon through a single experience, rather than situating that experience within a cultural context [Gor17]. For this study, it is particularly well-suited, since understanding slowability is not a matter of measuring features but of interpreting how they are encountered subjectively in practice.

As a Creative Technologist, I came to this study with a bias toward novel interface choices - something that made my preference for non-audio interactions stand out more strongly. To collect data, I recorded reflections immediately after engaging with each stop on the museum tour. In writing these notes, I kept the following guiding questions in mind:

1. What did I feel while using the app in this section?
2. Did time seem to slow down or speed up?
3. What interface elements captured my attention?

After completing the tour, I expanded on these notes through self-interviews, explicitly answering the above questions. I did notice, however, that my note-taking declined toward the end of the tour due to fatigue. To address this, I repeated the tour three times on separate days to ensure each stop received fair and balanced attention.

For analysis, I drew on Tishman's slow looking strategies as a framework [Tis18]. Applying these techniques allowed me to systematically interpret the experiential data and identify which app features fostered slowability, and which ones worked against it.

3.2. Case Context: The Jewish Museum Berlin and its Application

The Jewish Museum Berlin, which opened in 2001, serves as the site for this study. Designed by architect Daniel Libeskind, the building itself embodies an intentional and distinctive structure meant to evoke and recount German-Jewish history [Jü20]. The museum's core exhibition was updated in 2020, at which time a dedicated mobile application was launched to augment the visitor experience [Ayx20]. Rather than offering solely audio content, the app provides diverse multimedia formats including text, games, videos, and images, creating a comprehensive digital companion to the physical exhibition [Ayx20]. Central to the app's design philosophy is maintaining visitor autonomy - the app does not start playing content automatically, ensuring that "visitors decide for themselves how they want to use the app" [Ayx20]. This approach reinforces the principle that the app "enriches – but doesn't replace – the experience of touring the exhibition" [Ayx20].

When a user first launches the JMB app, it guides them through an initial setup process. They are first prompted to configure preferences for language, display contrast, text size, and content delivery (streaming or download). Following this, the app presents several modes for experiencing the museum: *Discover* (an independent exploration mode), *Highlights - Short Tour*, *Highlights - Long Tour*, *Architecture Gardens*, and information about the latest temporary exhibition [Jü20]. Figure 1 illustrates the key interface elements of the JMB app, showing the main tour selection screen and content delivery options.

To ensure a feasible and focused scope for the analysis, the Highlights- Long Tour was chosen. This tour, which consists of 29 stops, provides a comprehensive and structured path through the core exhibition, offering sufficient data points to analyze the app's features in detail [Jü20]. By focusing on this tour, this also allows for further research to be conducted using the same method, as auto-hermeneutics can set "precedents for further work" in exploring information experiences [Gor17].

The procedure at each of the 29 stops was consistent. First, the preliminary audio piece for the stop was played. Following the audio, the various interactive tabs displayed at the bottom of the screen were selected and explored. These tabs, which appeared up to two at a time, included [Jü20]:

- *Encounter* and *Expand*: Offered additional textual or visual information about the works.
- *Take Part*: Provided an accompanying interactive game or task.
- *Something Extra*: Contained supplementary details.
- *Listen*: Presented a musical interpretation of the stop.

3.3. Analysis: Features of Slowability in the JMB App

This section analyzes how the app fostered slowability through four of Tishman's strategies: categorization, open inventories, scope and scale, and juxtaposition. Each strategy created distinct affordances for slower, more deliberate engagement with the museum's exhibits and spaces.

Categorization

The app's categorization strategy was most evident at *Blindfolded* and *Voided Void*, where additional tabs created structured pathways for deeper engagement. These digital categories functioned as what Tishman describes as a "lens" [Tis18] that guided attention toward specific aspects of the experience.

At Blindfolded, the preliminary audio explained the two sculptures Ecclesia and Synagoga, providing historical context and identifying key features of each figure. As I recorded immediately after: "I liked being told exactly where to look, I was able to really focus on the details mentioned in the audio". This guided attention made me move closer to the sculptures and walk around them for proper examination. The subsequent Take Part tab presented six statue images, requiring me to identify which represented Synagoga. This interactive element transformed passive listening into active analysis, compelling me to re-examine the audio content and scrutinize each statue with heightened focus. My notes capture the satisfaction of this process: "I redid this exercise until I got them all right, there was something satisfying about completing it correctly".

Similarly, Voided Void layered multiple categories of experience. The preliminary audio established the Holocaust Tower's historical significance, while the Expand tab shifted my perspective to museum guide Jonas Hauer's spatial interpretation; Hauer, who is blind himself, offered an alternative way of experiencing the space [Ayx20]. The Something Extra tab added another layer, providing text that encouraged me to listen to how external sounds reverberate within the space. I documented the anticipatory effect: "Listening and reading this before entering made me curious as to what I'd discover in there." Once inside, the app's categories successfully directed my attention to acoustic phenomena I might otherwise have overlooked, creating what I noted as "a sense of satisfaction when I picked up on what was mentioned".

However, categorization did not always enhance slowability. At the stop *Kashrut*, the Take Part tab's drag-and-drop interface created technical friction that disrupted rather than facilitated engagement. This represented what Björneborn would classify as unproductive rather than generative stumbling [Bjö17]—a barrier that threatened to pull me out of the immersive experience entirely.

Open inventories

The *Libeskind Building* stop exemplified Tishman's open inventory strategy, deliberately withholding immediate interpretation, allowing for an unmediated encounter with the space [Tis18]. As the tour's opening point, architect Daniel Libeskind's recorded introduction explicitly challenged conventional audio guide practices: "audio guides usually tell you right away something, but I think it would be good to give people a chance not to know where they are, to let them experience something in silence" [Jü20].

This invitation "not to know" prompted me to remove my headphones and engage in unguided observation. My immediate response, as recorded: "Everything here is at angles—this makes me feel uneasy." I noticed the stark aesthetic—high ceilings, white walls, angular geometry—while distant drumming from another exhibition contributed to what I documented as feelings of unease and anxiety.

This open inventory approach fundamentally altered my relationship to the space. Rather than receiving predetermined information, I was positioned as an active interpreter, attending to whatever captured my attention. As I noted: "Without being told what to notice, I found myself looking at the architecture in a way I normally wouldn't." This self-directed analysis slowed my movement through the space and heightened my awareness of environmental details that might otherwise have remained peripheral.

Scope and Scale

The app's scope and scale manipulations were most sophisticated in *People in Pictures*, *Manheimer Family Portrait*, and *Joseph Süss Oppenheimer*. All three stops employed the use of zooming, which, as Tishman describes helps, "bring certain features into relief" [Tis18].

The first two stops featured Encounter tabs with interactive white

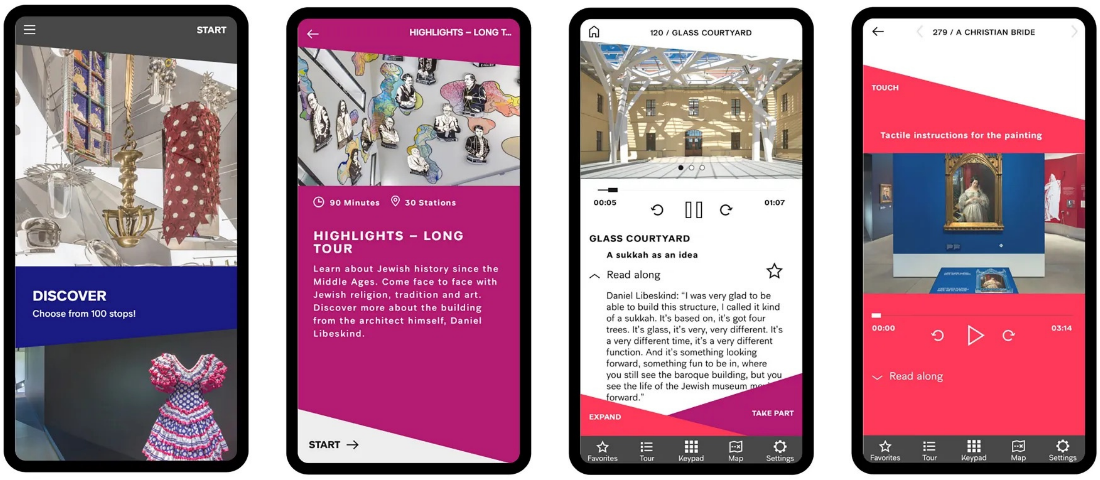

Figure 1: *Screenshots of the JMB App interface showing: (left) the 'Discover' independent exploration mode, (center-left) the 'Highlights - Long Tour' selection, (center-right) interactive content tabs at the bottom of the screen, and (right) an example of a pure audio stop interface [Ayx20].*

circles scattered across painting reproductions. Selecting these circles zoomed into specific areas while accompanying audio provided contextual information. I recorded my initial reaction: "These circles make me feel intrigued- I want to see why this part is deemed important." This gamified approach to visual analysis created curiosity and investment in paintings that might otherwise have been overlooked.

The zooming function proved more impactful than the accompanying audio. When examining magnified sections, I found myself comparing digital details with the physical artwork, leading to observations about brushwork, lighting techniques, and compositional elements. My notes capture this comparative process: "Looking back and forth between the zoomed image and the real painting, I start noticing things about the brushstrokes and lighting that I might have just walked past." This digital-physical dialogue necessitated closer physical proximity to the artworks and extended my engagement time significantly.

The Joseph Süss Oppenheimer exhibit demonstrated scope and scale's effectiveness with complex artifacts. The medallion depicting Oppenheimer's life contained numerous small images that were difficult to interpret without magnification. The accompanying video and audio narrated his biography while highlighting specific visual elements, providing what Tishman describes as a "structure for visitors to go beyond a first glance" [Tis18] and what I documented as sustained attention: "Instead of just glancing at this and moving on, I actually stayed to understand the whole story."

Juxtaposition

Juxtaposition emerged most powerfully in the app's presentation of alternative perspectives on Liebeskind's architectural spaces. As Tishman explains, juxtaposition works by "placing objects next to each other with the purpose of bringing forward certain features through comparison" [Tis18]. Voided Void, *Garden of Exile*, and Libeskind Building each offered contrasting sensory interpretations that challenged conventional spatial experience.

The most striking juxtaposition occurred in Garden of Exile, where the app presented both sighted and Hauer's perspective as a blind visitor on the deliberately disorienting space. While the uneven floor and angled columns created visual unease, Hauer described the space as relatively calm and navigable. To test this perspective, I closed my eyes and tried his approach: "It actually does feel calmer when you're not seeing everything at an angle." This perceptual shift revealed tactile qualities of the columns and spatial relationships I had initially overlooked.

The Listen tabs in both Garden of Exile and Libeskind Building provided musical interpretations that amplified the spaces' intended emotional effects. The Garden of Exile's score—which I described as "water draining in an industrial space with occasional screechy wind instruments"—enhanced the disorientation, while Libeskind Building's haphazard wind instruments created what I documented as unease and stress. These sonic interpretations functioned as affective translations of Libeskind's architectural intentions, giving further insight into what the architect was trying to convey through the spaces.

Through these four strategies, the JMB app successfully created multiple affordances for slowability, transforming what might have been a conventional museum visit into an extended, multi-sensory investigation of space, artifact, and perspective. The most effective implementations combined digital innovation with phenomenological awareness, creating productive friction that encouraged deeper engagement rather than mere efficiency.

4. Discussion

The JMB app case study reveals the complex dynamics between guided attention and open exploration in fostering slowability, illuminating fundamental tensions that extend beyond museum design to broader questions of attention management in digital environments.

The Paradox of Guided Attention

Both categorization and scope and scale strategies demonstrated clear effectiveness in directing focus and extending engagement time with exhibits. However, these guided approaches simultaneously can reduce the occurrence for users to find new things, as Björneborn explains, "serendipity may happen in environments with incomplete, inconsistent, and 'unfinalizable' features that leave potentials open to us" [Bjö17]. At Blindfolded, while the audio successfully guided me to examine specific iconographic features of Ecclesia and Synagoga, this predetermined focus prevented me from giving attention to other sculptural qualities—facial expressions, the treatment of fabric in marble, or spatial relationships between the figures. Similarly, in Voided Void, my attention narrowed to acoustic phenomena at the expense of other sensory dimensions like temperature or smell that might have enhanced my spatial understanding.

This selective attention paradox extends to the painting encounters in People in Pictures and Manheimer Family Portrait, where interactive white circles directed focus to curator-selected details while bypassing my usual analytical frameworks around brushwork and compositional structure. As Tishman observes, "exhibits seem designed to make an argument rather than invite speculation" [Tis18], creating what I experienced as productive yet constraining guidance that enhanced depth while limiting breadth of discovery.

These findings align with Björneborn's concept of environmental "incompleteness" [Bjö17], which he argues is essential for serendipitous discovery. When digital interfaces predetermine significant features through categorization strategies, they constrain interpretive possibilities. As Tishman observes, "The categories we use to focus our attention profoundly shape what we see. They also shape what we think" [Tis18].

Open Inventories as Counterbalance

The Libeskind Building stop provided a compelling counterexample through its open inventory approach. Libeskind's explicit instruction to experience the space "in silence" before receiving interpretation created what Björneborn identifies as conditions for discovering, "things we do not know we do not know: the unknown unknowns or yet-to-be-knowns" [Bjö17]. This unguided approach enabled environmental incompleteness—rather than closing down interpretive possibilities, the app preserved what I documented as opportunities for personal discovery, including integration of distant drumming and awareness of how angular geometry affected my physical comfort.

The open inventory strategy demonstrated how effective slowability design can create structured serendipity that slows attention without predetermining discovery outcomes, allowing for what Tishman claims to, "capture the rich, often category-defying jumble of features that make up a whole" [Tis18], fostering discoveries that transcend predetermined organizational frameworks.

Productive Juxtaposition and Cross-Contacts

The app's juxtaposition strategy, particularly evident in Voided Void and Garden of Exile, demonstrated how contrasting perspectives can enhance rather than constrain experiential possibilities. Hauer's spatial descriptions functioned as what Björneborn terms "cross-contacts" [Bjö17], which are, "moments when dissimilar resources (information, things, people, etc.) meet or collide across contact surfaces, edges, intersections" [Bjö17]. This digital cross-contact operated precisely as Tishman describes effective juxtaposition: by placing my sighted experience alongside Hauer's tactile and auditory interpretation, the app demonstrated how juxtaposition can be "strategically used to draw attention to subtle differences" [Tis18]—in this case, between two fundamentally different modes of encountering architectural space.

Slow Looking as Cultural Counter-Skill

These design tensions reflect broader challenges within what scholars term the attention economy, where digital media compete to capture and hold users' focus through immediacy-driven features [GLAAFR20]. Against this backdrop, slow looking functions as what Tishman identifies as "an important counterbalance to the natural human tendency toward fast looking" [Tis18]. In digital contexts, this counterbalance becomes essential as our default mode involves swift, unreflective engagement with visual content—absorbing only immediate, accessible information before rapidly shifting attention elsewhere.

Museums represent ideal environments for cultivating sustained attention. As cultural heritage institutions designed for contemplation, they can serve as practice rounds for deep engagement skills that have broader applications beyond museum walls [Hei12]. However, as this case study demonstrates, simply providing digital tools is insufficient; the design of these tools fundamentally shapes whether they support or undermine prolonged attention.

The auto-hermeneutic findings suggest that effective slowability design requires careful calibration between guidance and openness. The most successful app features preserved what Björneborn identifies as essential: space for individual discovery while providing sufficient structure to encourage sustained engagement [Bjö17]. As Tishman notes, visitors often judge museum quality not by information acquisition but by whether the experience increased their curiosity [Tis18]—a metric that depends precisely on maintaining the delicate balance between structured guidance and serendipitous discovery that this study reveals.

5. Conclusion

The attention economy has left its mark on museums, where visitors often spend only fleeting moments with exhibition pieces. While group-based initiatives such as Slow Art Day have attempted to counter this trend, little attention has been given to how individual digital experiences might foster the same sustained engagement. This paper suggests that one way forward lies in designing museum applications with slowability in mind.

The analysis of the JMB app showed how particular strategies created conditions for slow looking. Open inventory and juxtaposition encouraged reflection by combining direction with autonomy, allowing visitors to notice details that resonated personally while still being guided by the app. These features were most effective when they expanded rather than narrowed perception, often through audio narration and giving alternative perspectives. By contrast, categorization and scale/scope sometimes risked over-directing attention, pointing visitors to specific features that left little room for serendipitous discovery.

As an auto-hermeneutic case study, this research does not aim to generalize but to highlight the importance of designing for sustained attention [Gor17]. Future studies, whether at the JMB or in other institutions, could build on these findings to identify more concrete design principles.

In an era where digital tools are often designed for speed and distraction, museums have the opportunity to move in the opposite direction. By intentionally designing for slowability, they can help visitors rediscover the practice of sustained attention—an essential skill not only for engaging with art and history, but for navigating contemporary life.

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